

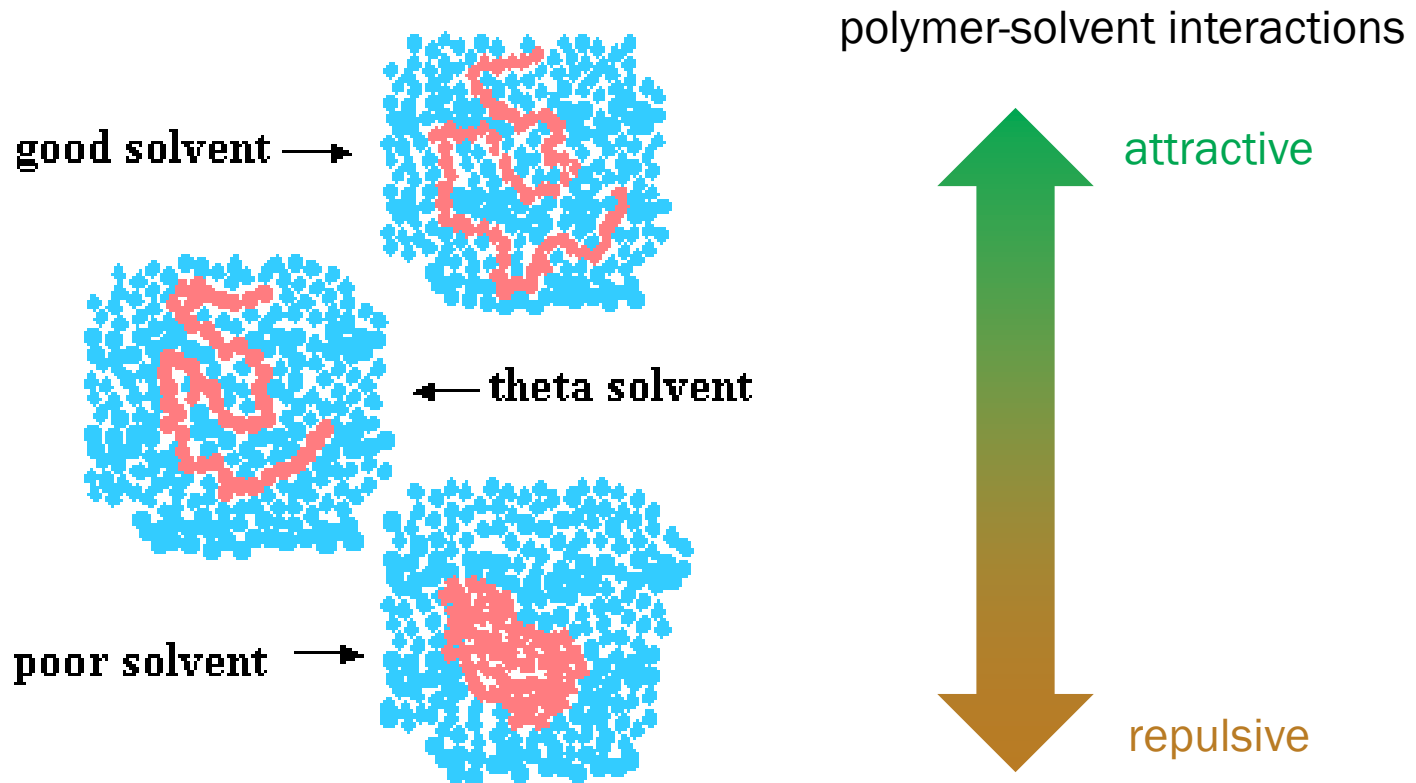
Physical Chemistry of Materials

Lecture 21

Branching and Gelation



Long-Range Interactions



(Image: http://rkt.chem.ox.ac.uk/lectures/liqsolns/polymer_solutions.html)

Flory Exponent

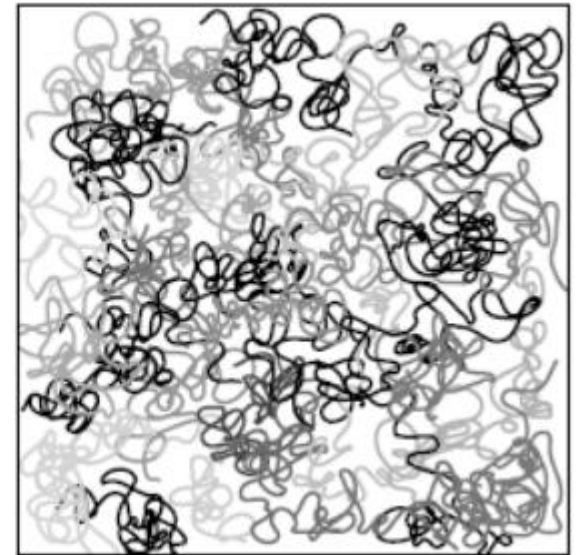
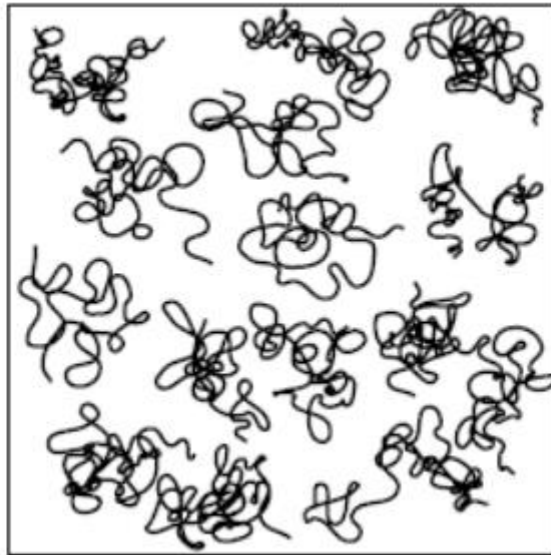
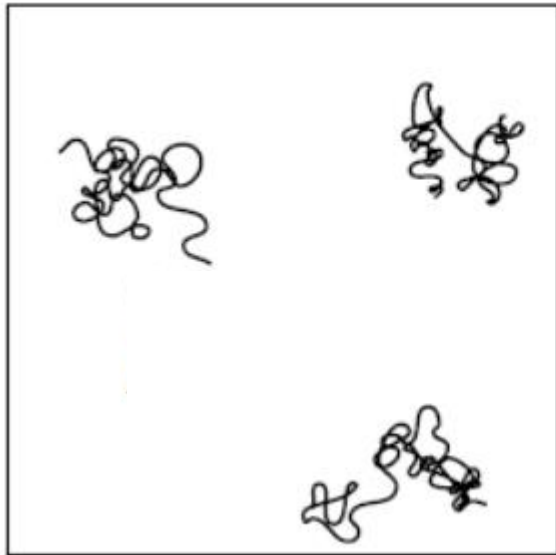
Generally, $R \propto N^\nu$

- Ideal chain (theta solvent): $\nu = 0.5$
- Good solvent: $\nu > 0.5$
 - Flory's prediction: $R_{\text{good}} = v_0^{1/5} a^{2/5} N^{3/5}$
- Poor solvent: $\nu < 0.5$
 - Flory's prediction: $R_{\text{poor}} = (w/|v_0|)^{1/3} N^{1/3}$

Last Class

Polymer Solutions

polymer concentration



“overlap”

Theta Solvent

$$R \sim a\sqrt{N} \text{ (at all concentrations)}$$

- Overlap volume fraction ϕ^*

(system volume fraction) = (volume fraction in a coil)

$$\phi^* = \frac{Na^3}{R^3} = \frac{1}{\sqrt{N}}$$

- Dilute solution: $\phi < \phi^*$
- Semidilute solution: $\phi > \phi^*$ (but still $\phi \ll 1$)

Good Solvent

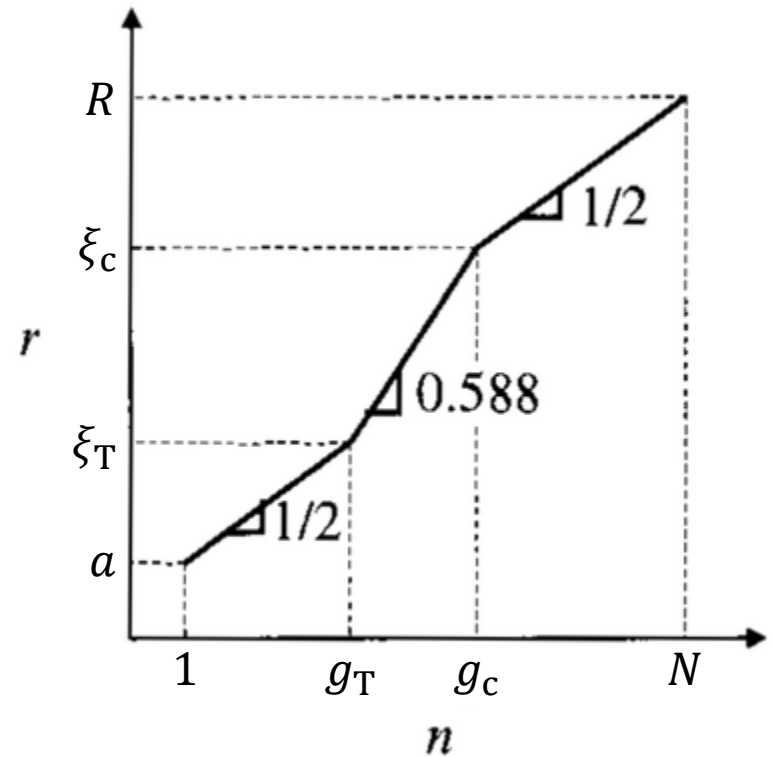
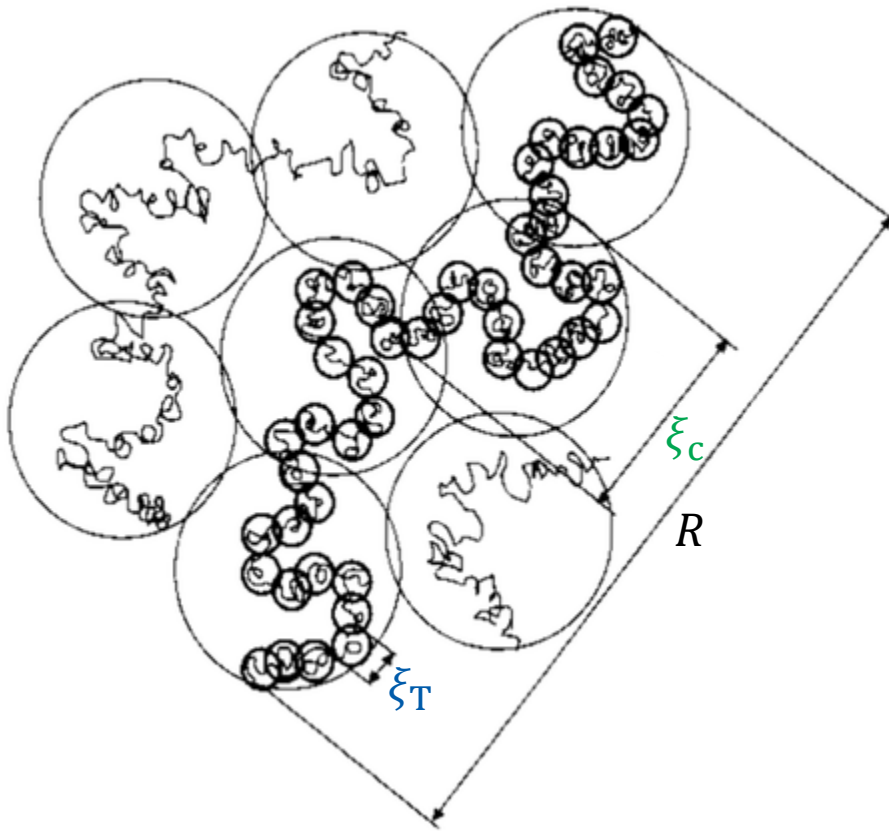
$$R_{\text{good}} \approx v_0^{2\nu-1} a^{4-6\nu} N^\nu$$

- Overlap volume fraction ϕ^*

$$\phi^* = \frac{Na^3}{R_{\text{good}}^3} = \left(\frac{a^3}{v_0}\right)^{6\nu-3} N^{1-3\nu} \sim N^{-0.76}$$

What happens if $\phi > \phi^*$?

Three Regimes



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 5.5)

Concentrated Solution

$$\xi_T \approx \frac{a^4}{v_0}; \quad \xi_c \approx a \left(\frac{a^3}{v_0} \right)^{(2\nu-1)/(3\nu-1)} \phi^{-\nu/(3\nu-1)}$$

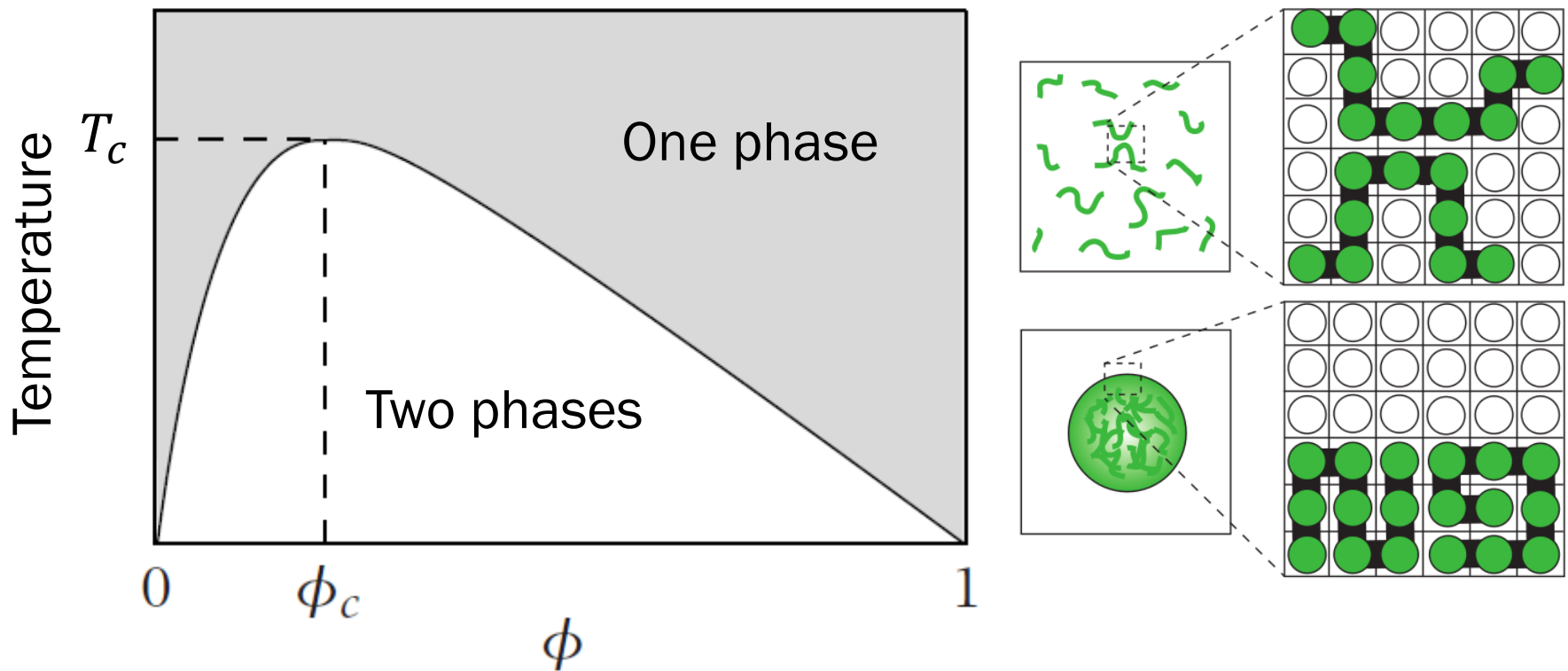
The crossover point is

$$\phi^{**} = \frac{v_0}{a^3}$$

For $\phi > \phi^{**}$, chains are **ideal on all length scales**.

→ concentrated solution

Poor Solvent



(Images: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figure 32.7;
Brangwynne et al., *Nature Physics* (2015))

Flory-Huggins Model

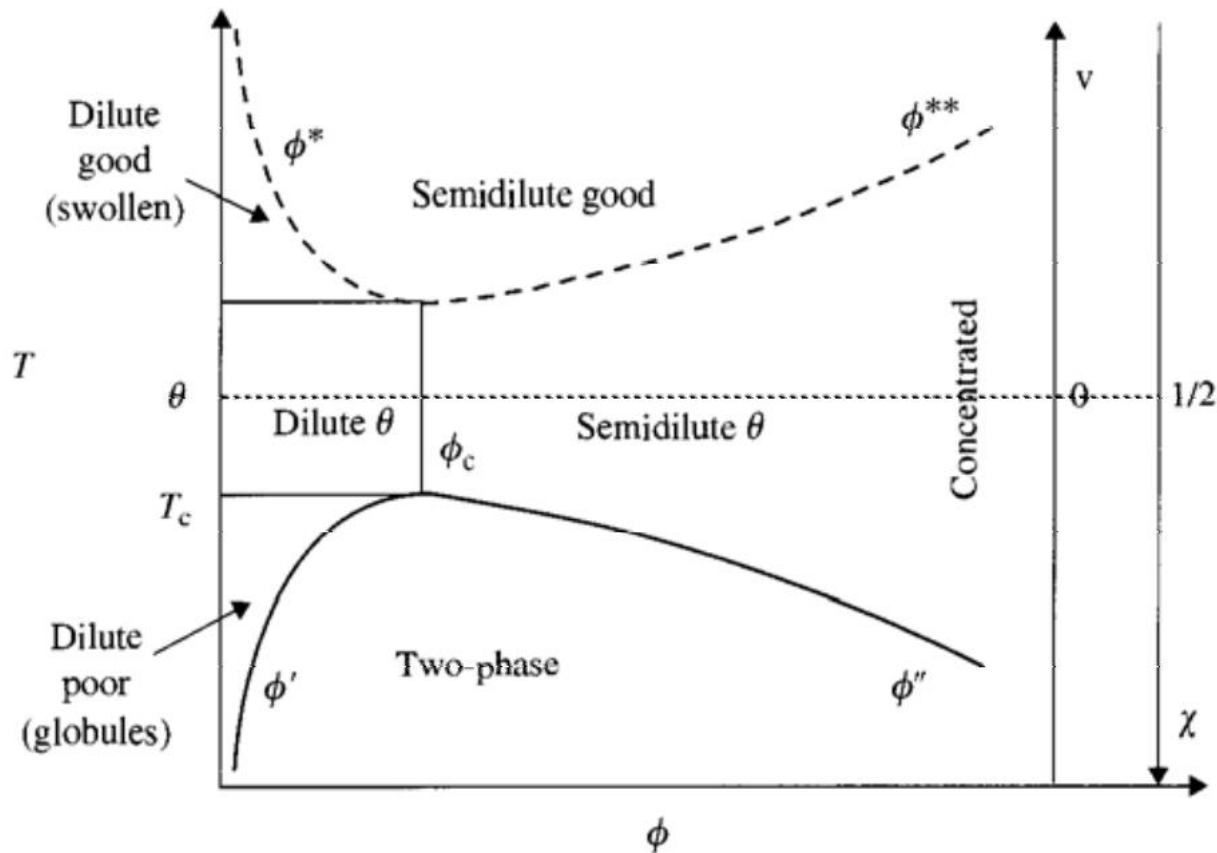
$$\Delta \bar{S}_{\text{mix}} = -k_B \left(\frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B \right)$$

$$\Delta \bar{U}_{\text{mix}} = \chi k_B T \phi_A \phi_B$$

$$\Delta \bar{A}_{\text{mix}} = \Delta \bar{U}_{\text{mix}} - T \Delta \bar{S}_{\text{mix}}$$

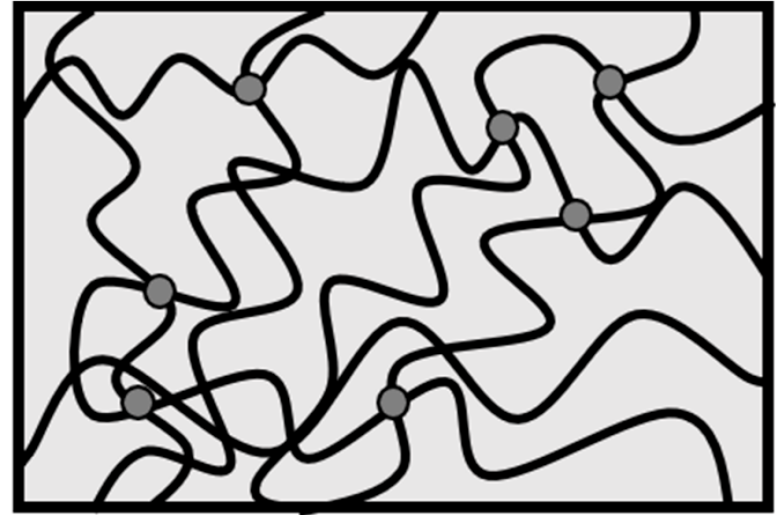
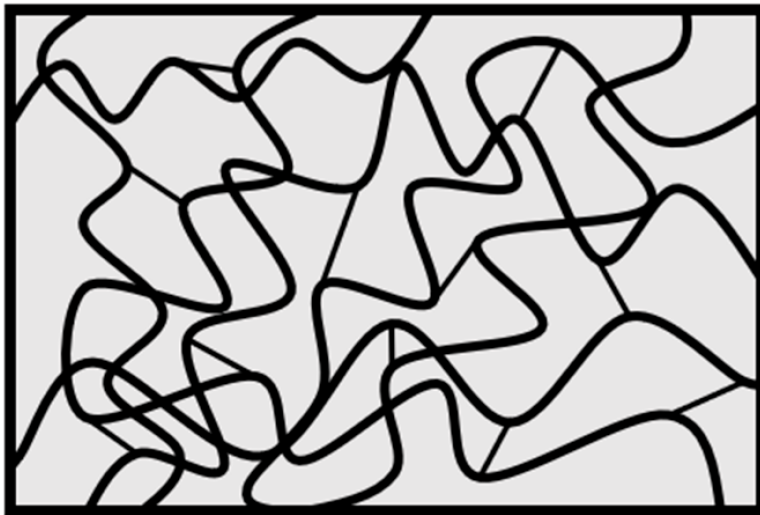
$$\Delta \bar{A}_{\text{mix}} = k_B T \left(\frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B + \chi \phi_A \phi_B \right)$$

Summary



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 5.1)

Branching and Gelation



Polymer chain

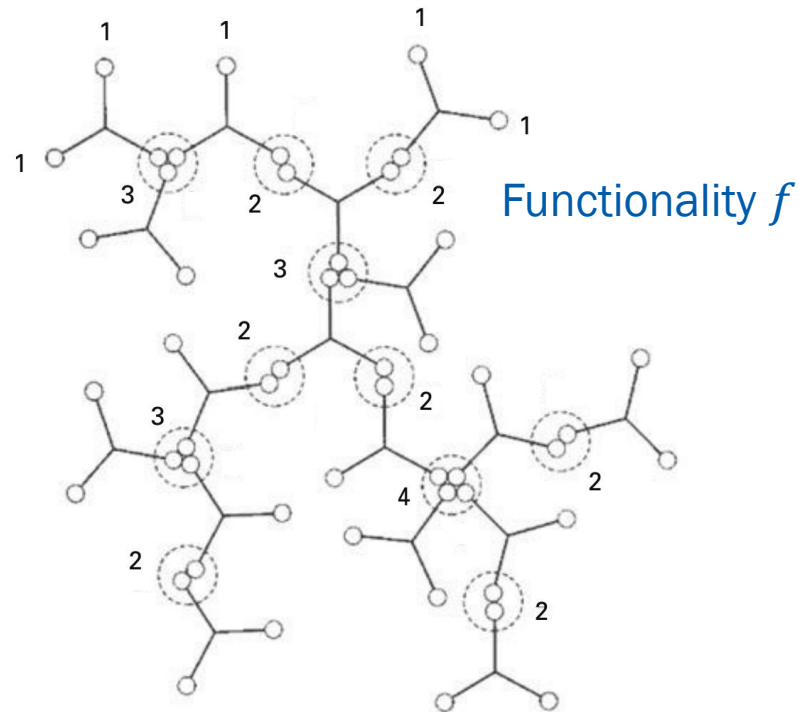


Chemical crosslink



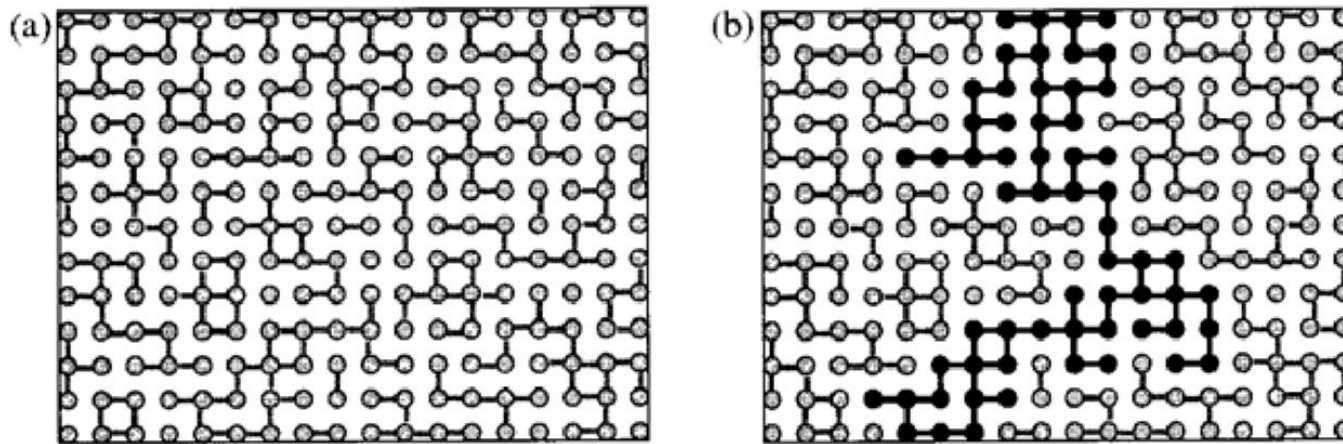
Physical crosslink

Network Structure



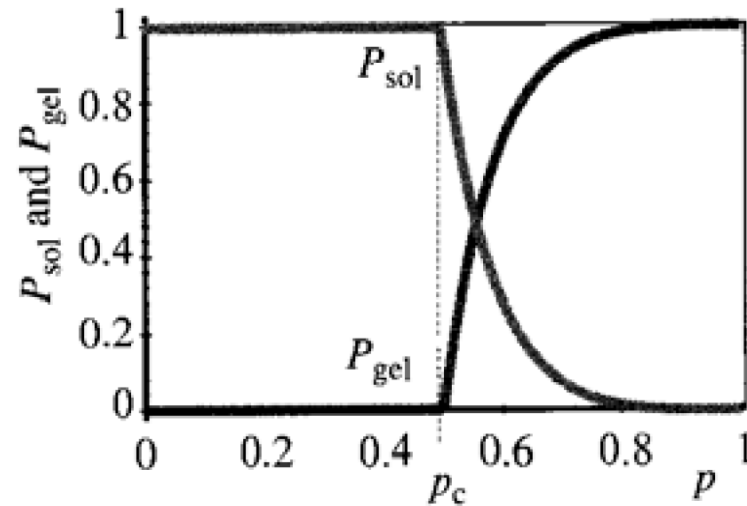
(Image: Tanaka, *Polymer Physics* (2011), Fig. 3.3)

Percolation: Network Formation



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 6.14)

Sol-Gel Transition



p = extent of reaction

P_{sol} = “sol” fraction

P_{gel} = “gel” fraction

p_c = gel point

1-Dimensional Model

... AB AB-AB AB-AB-AB AB AB AB-AB ...

- Extent of reaction p (= bond formation probability)
- For $p < 1$, the system only contains finite polymers
- For $p = 1$, the system contains one infinite polymer (“percolation”)
- Gel point: $p_c = 1$

1-Dimensional Model

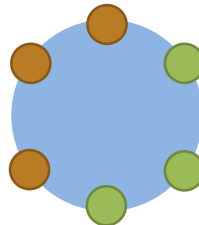
... AB AB-AB AB-AB-AB AB AB AB-AB ...

- Extent of reaction p (= bond formation probability)
- Number of monomers n
- Number of polymers $n - np$
- Average degree of polymerization

$$N_n(p) = \frac{1}{1 - p}$$

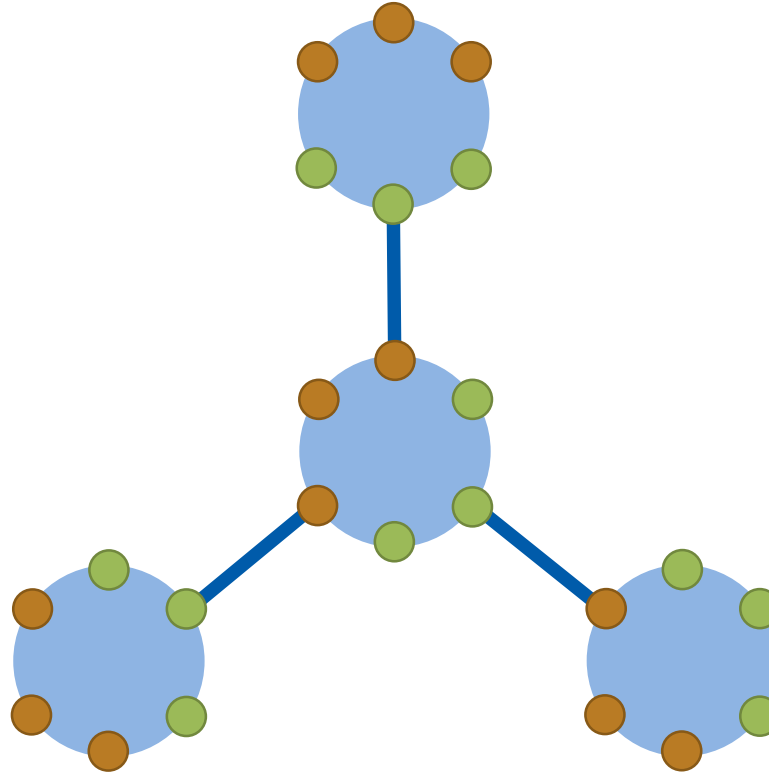
Flory-Stockmayer Theory

- Developed by Flory (1941) and Stockmayer (1944)
- A monomer with **functionality f** and **extent of reaction p**

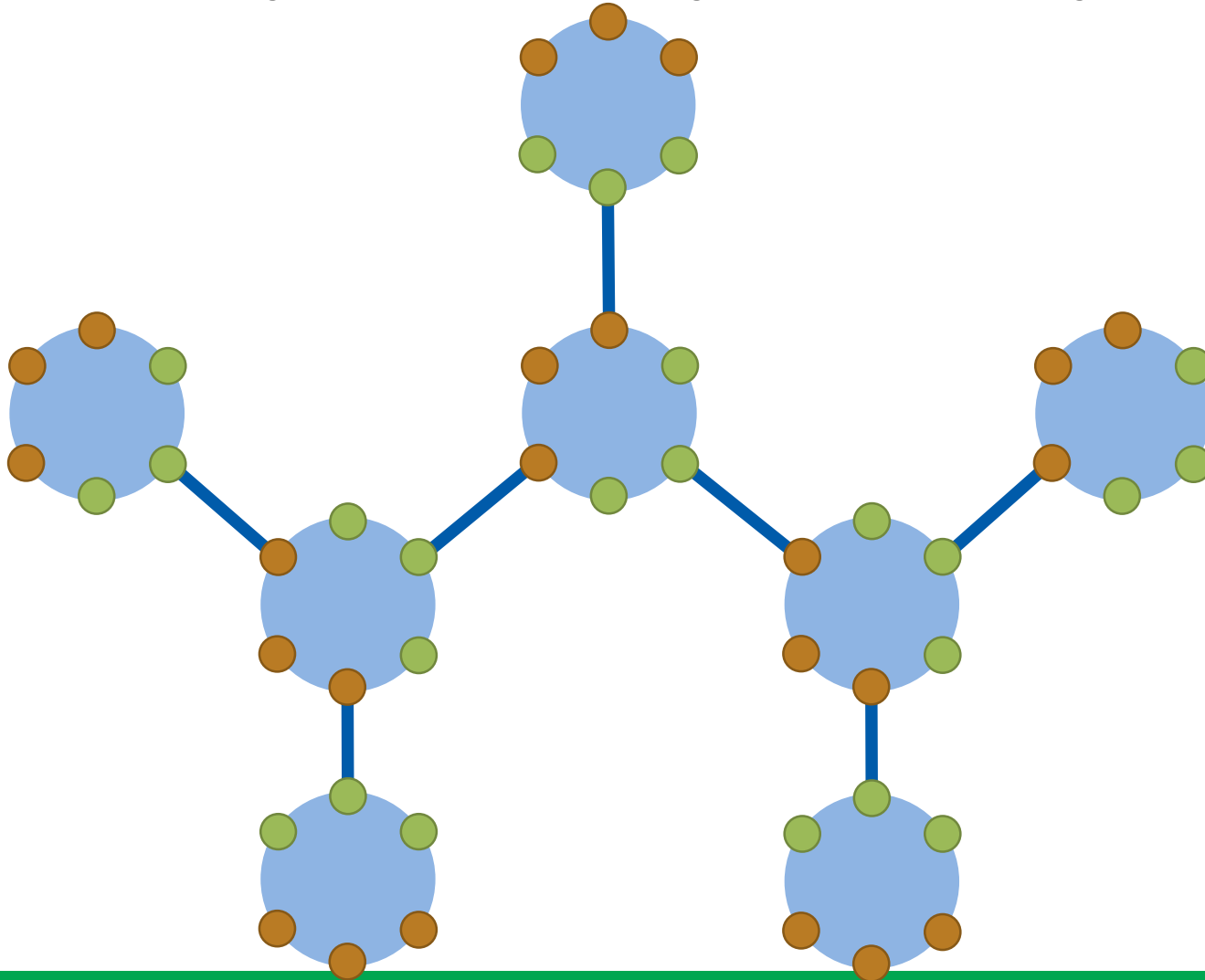


- Construction of a network by probabilistic linkage
- Classical, mean-field theory of gelation

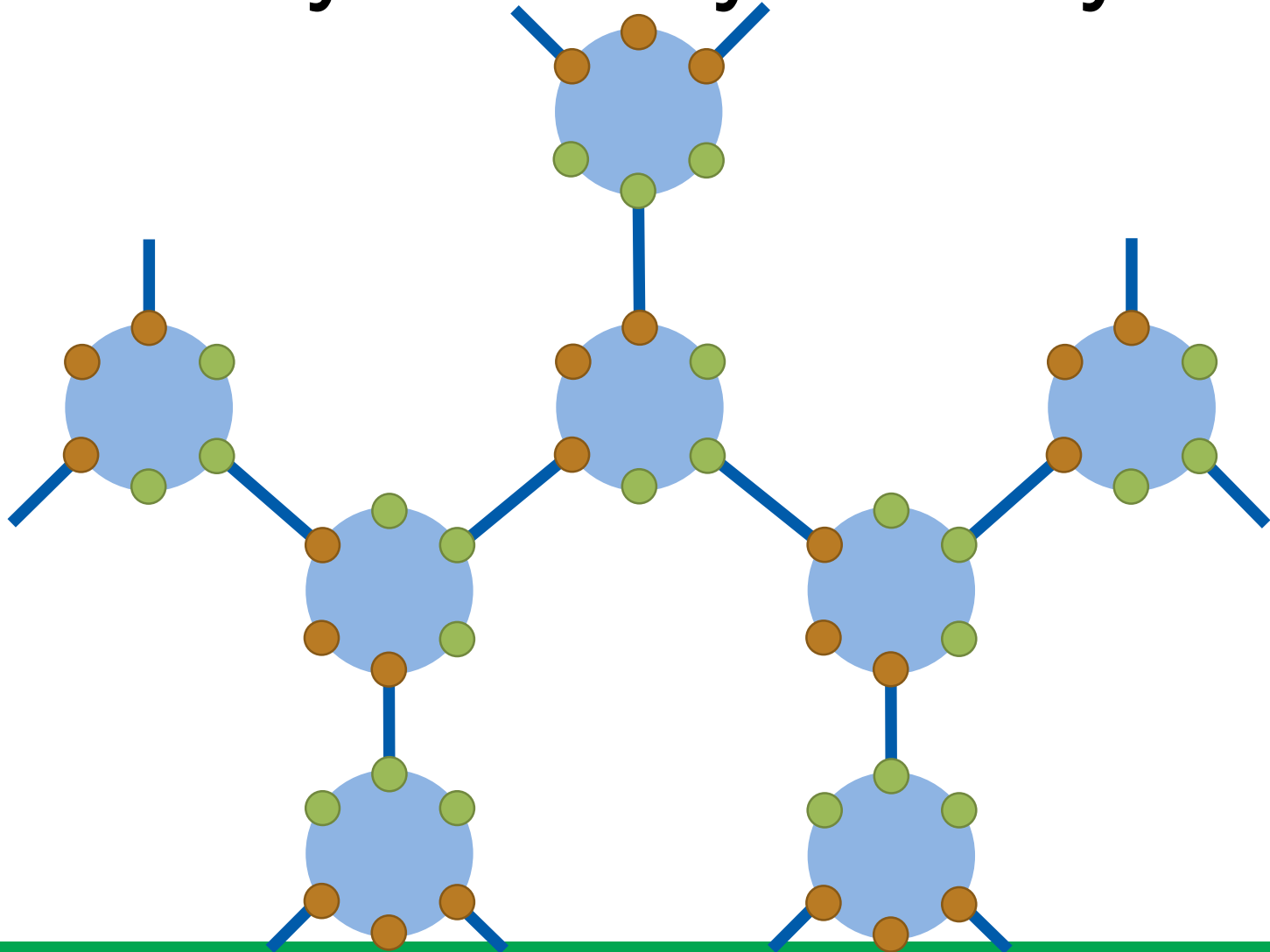
Flory-Stockmayer Theory



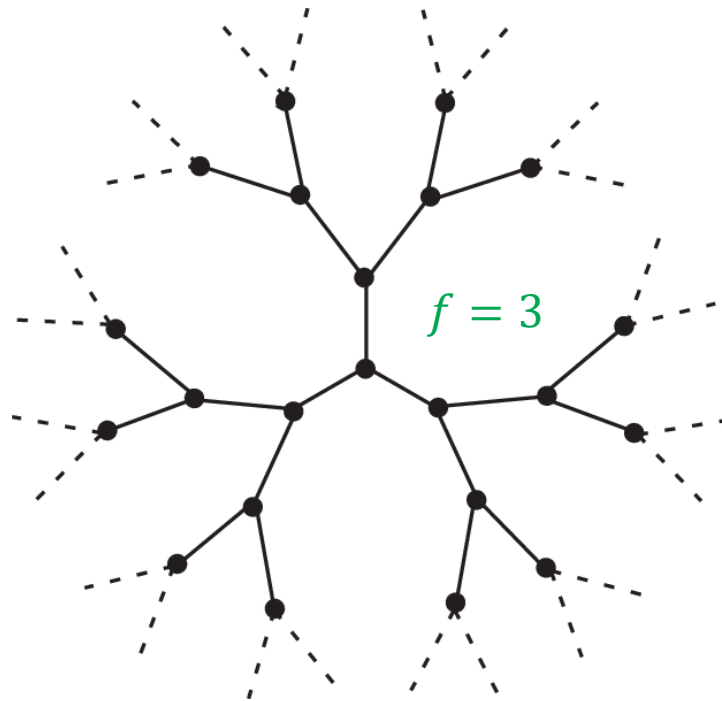
Flory-Stockmayer Theory



Flory-Stockmayer Theory



Bethe Lattice (Cayley Tree)



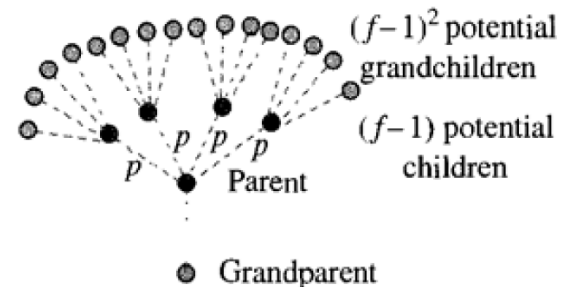
(Image: ben-Avraham and Glasser, *J Phys Condens Matter* (2007), Fig. 1)

Gel Point

- Extent of reaction p (= bond formation probability)
- Average number of bonds $p(f - 1)$
- Gel point: network propagation

$$p(f - 1) > 1$$

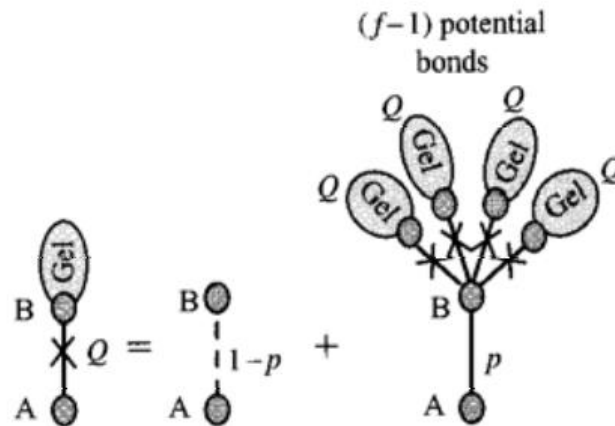
$$p_c = \frac{1}{f - 1}$$



Sol and Gel Fractions

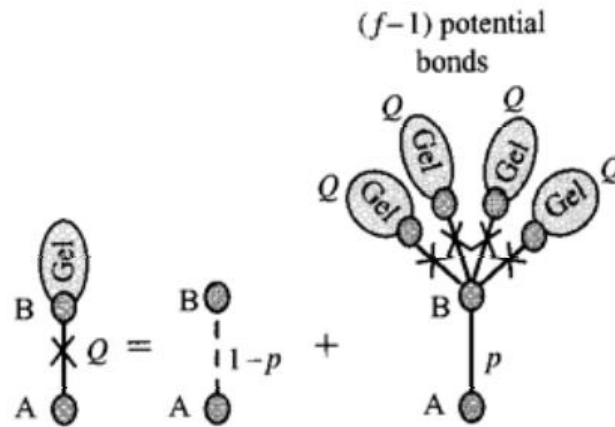
Using the same model, one can obtain **sol and gel fractions**.

Define Q as the probability that a randomly selected site is not connected to the gel.



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 6.17)

Sol and Gel Fractions



$$Q = 1 - p + pQ^{f-1}$$

However, for monomers in the sol,

$$P_{\text{sol}} = Q^f$$

Sol and Gel Fractions

$$Q = 1 - p + pQ^{f-1}$$

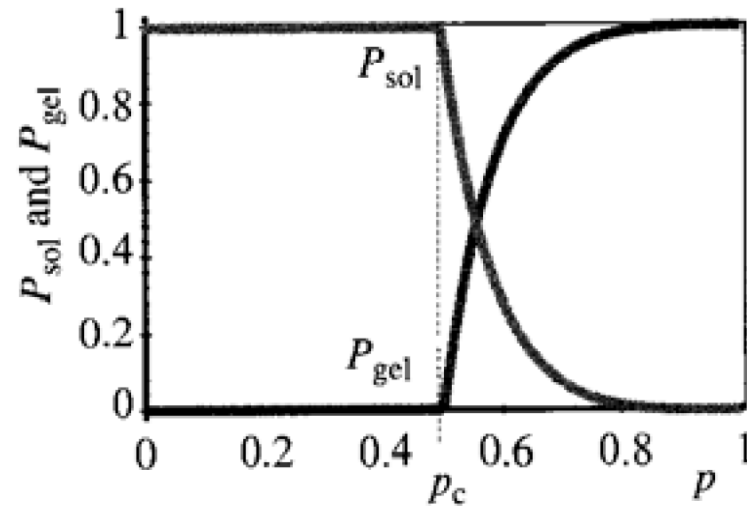
$$P_{\text{sol}} = Q^f$$

$$P_{\text{sol}}^{1/f} = 1 - p + pP_{\text{sol}}^{(f-1)/f}$$

- This equation always has one solution $P_{\text{sol}} = 1$
- For $f \geq 3$, there is another solution $P_{\text{sol}} < 1$ (“gel”)

e.g. $f = 3$: $P_{\text{sol}} = (1 - p)^3 / p^3$ for $p > p_c$

Sol-Gel Transition



p = extent of reaction

P_{sol} = “sol” fraction

P_{gel} = “gel” fraction

p_c = gel point

Average Degree of Polymerization

- Assume that $p \leq p_c$ (before the gel point)
- Note: no intramolecular bonds in the Bethe lattice
- Maximum number of bonds per monomer = $f/2$
- Total number of formed bonds per monomer = $pf/2$
- Average number of polymers per monomer = $1 - pf/2$
- Average number of monomers per polymer

$$N_n(p) = \frac{1}{1 - pf/2}$$

At the Gel Point

- Average number of monomers per polymer

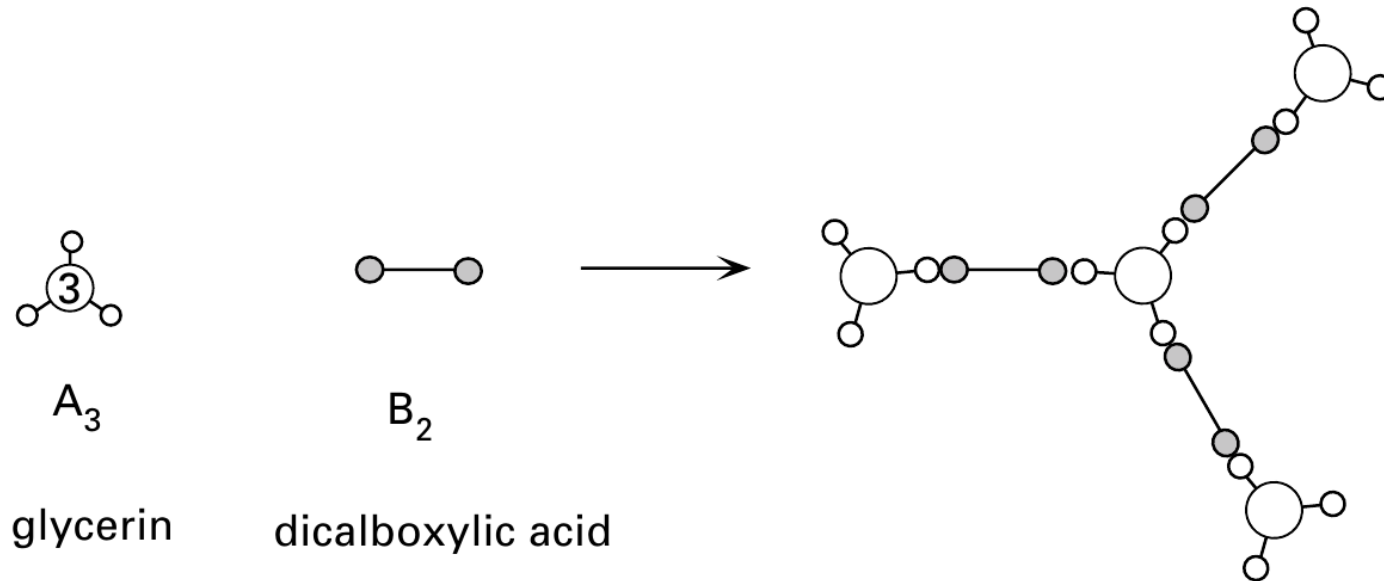
$$N_n(p) = \frac{1}{1 - pf/2}$$

- At the gel point, $p_c = 1/(f - 1)$ and

$$N_n(p_c) = \frac{1}{1 - f/2(f - 1)} = \frac{2(f - 1)}{f - 2}$$

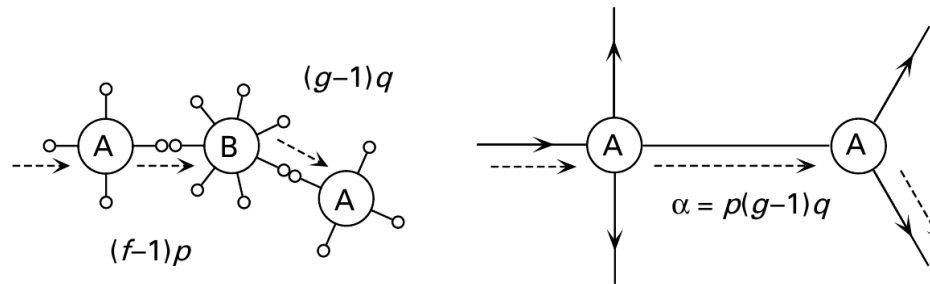
e.g. $f = 3$: $N_n(p_c) = 4$ (NOT so large!)

Heteromolecular Condensation



Gel Point

- Functionalities: f for A and g for B
- Extents of reaction: p for A and q for B



- Gel point: $p(f-1)q(g-1) = 1$

Revisit the Gel Point

For $f = 3$,

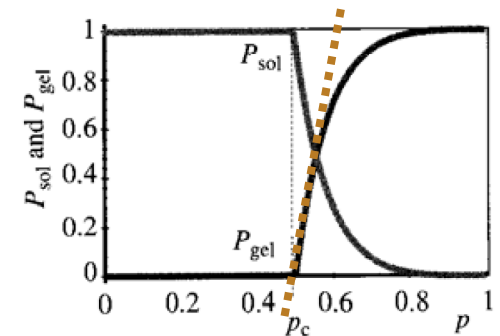
$$p_c = 1/(f - 1) = 1/2$$

$$P_{\text{sol}} = (1 - p)^3/p^3$$

$$P_{\text{gel}} = 1 - (1 - p)^3/p^3$$

Define $\varepsilon = (p - p_c)/p_c$ and one can show that

$$P_{\text{gel}} \approx 6\varepsilon \propto (p - p_c)^1$$



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 6.15)

Critical Exponent

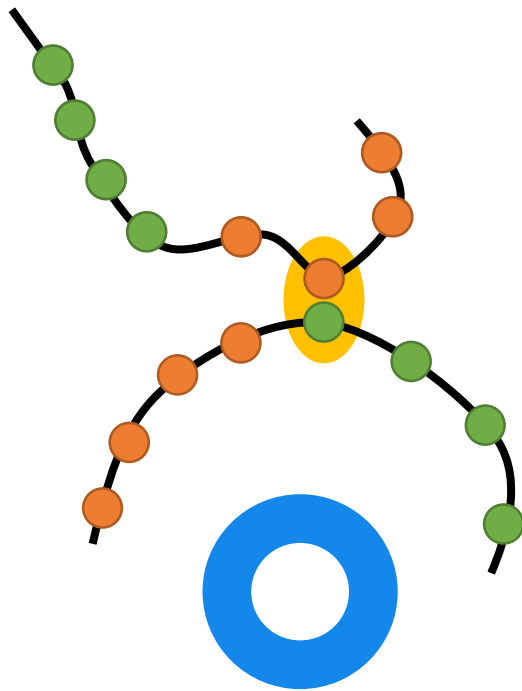
$$P_{\text{gel}} \sim (p - p_c)^\beta$$

- Flory-Stockmayer theory: $\beta = 1$ (works for $D \geq 6$)
- Computer simulation for 3D system: $\beta \approx 0.41$

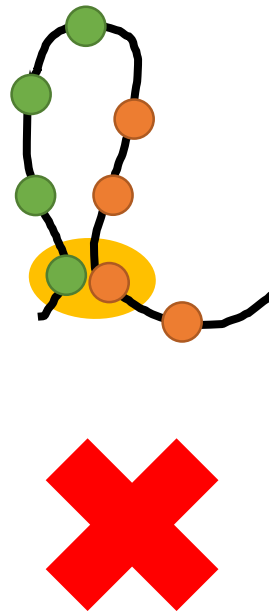
Why does it fail?

Assumption in Mean-Field Model

One-bond inter-chain interactions



Intra-chain interactions



Multi-bond interactions

